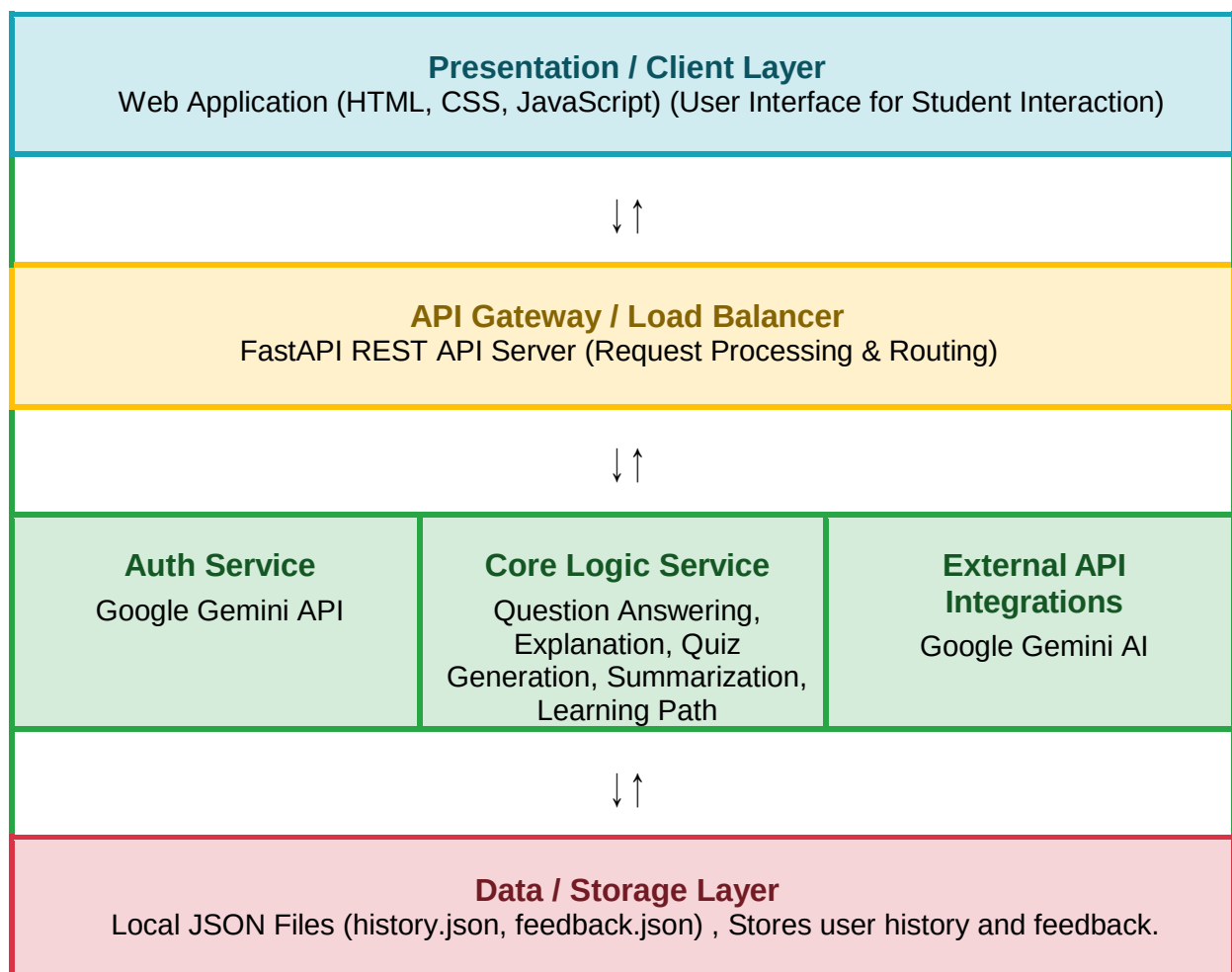


Solution Architecture

Date	28 JUNE 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface where students can ask questions, get explanations, summarize text, generate quizzes, and create learning paths.	HTML, CSS, JavaScript
API Gateway	Receives requests from the user interface, validates inputs, and routes them to the appropriate AI module.	FastAPI, Uvicorn
AI Services / Core Logic	Processes user requests such as question answering, explanation generation, summarization, quiz generation, and personalized learning paths using AI.	Python, Google Gemini API
Database / Storage	Stores conversation history and user feedback locally for future reference	JSON Files (history.json, feedback.json)
External API Integration	Connects the application with Google's Gemini AI model to generate intelligent educational responses	Google Gemini API
Testing Layer	Tests API endpoints and verifies application functionality.	PyTest